Activity 2

Activity Name: Automatic Light Control System





The Problem: One evening, Rohan was studying in his room when the sunlight outside became dim. He noticed that the room was getting dark, but the light was still OFF. He switched ON the light manually but later wondered, "Why should I turn the light ON and OFF every time the brightness changes?" Rohan wanted to create a system that could detect the amount of light in the room and automatically control the light.

What Will We Build: The students will be able to build an automatic light control system that senses how bright or dark the room is. The LDR will detect the light, and the Arduino will automatically turn the LED ON or OFF.

Components Required





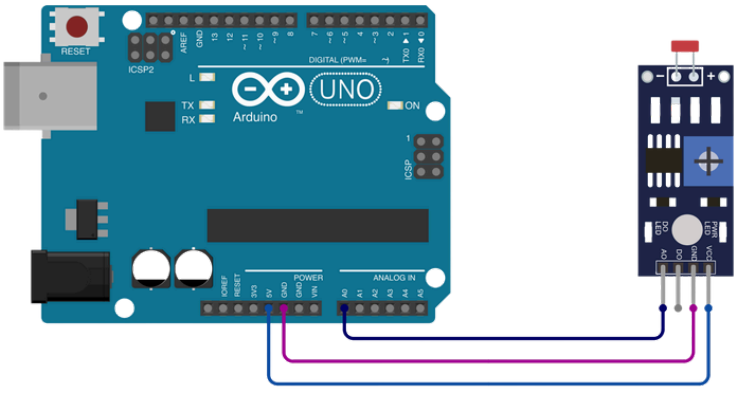




Circuit Designing

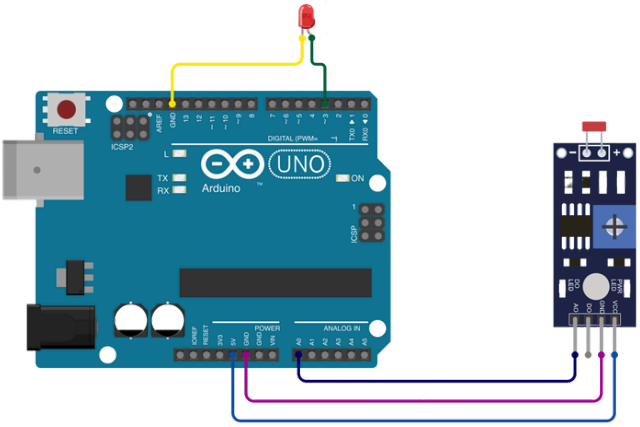
Step 1: Connect Arduino with the LDR Sensor.

Sensor Pins	Arduino Pins
Vcc Pin	5V Pin
GND Pin	GND Pin
A0 Pin	Analog A0 Pin



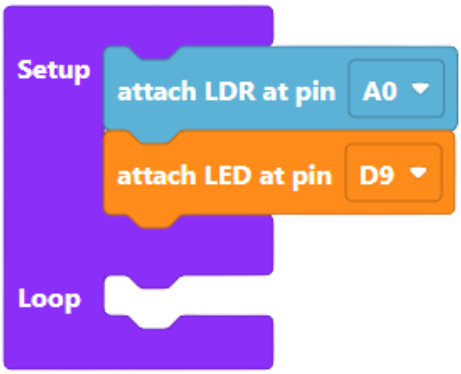
Step 2: Connect the LEDs' to the Arduino Uno.

LED Pins	Arduino Pins
Positive Pin	Digital Pin 4
Negative Pin	GND Pin

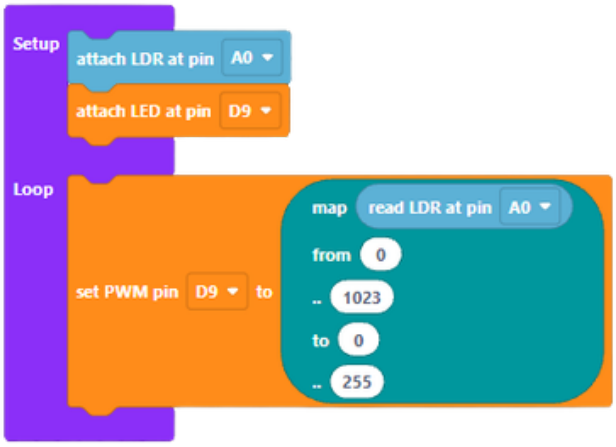


Project Coding

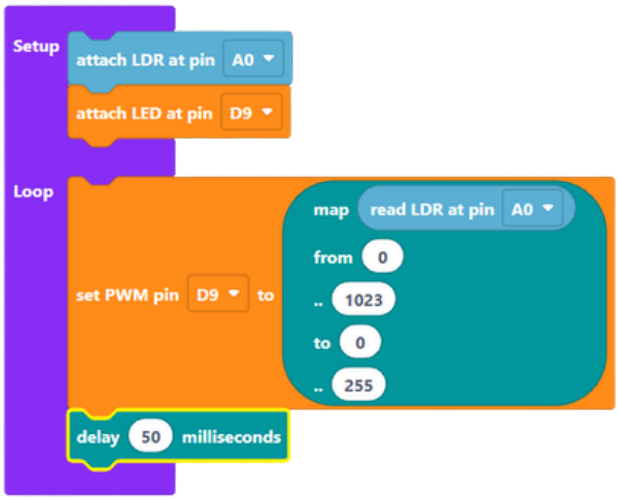
Step 1: Make code to initialize the pins



Step 2: Make code to set the light intensity of the LDR sensor



Step 3: Make code to use the delay for the time duration.



Learning Outcomes

- The students will be able to identify the LDR, Arduino Uno, jumper wires, and LED used in the project.
- The students will be able to connect the LDR and LED to the Arduino correctly.
- The students will be able to understand how an LDR detects light and darkness.
- The students will be able to create simple code to control the LED based on the light level.



BRAIN TEASERS

Section A – Multiple Choice Questions

1. Which component detects the amount of light?

A. LED
B. LDR
C. Jumper Wire
D. Arduino Uno

2. Which board controls the LED in this project?

A. Arduino Uno
B. Keyboard
C. Speaker
D. Mouse

3. What happens when the room becomes dark?

A. The LED can turn ON
B. The Arduino turns OFF
C. The LDR disappears
D. The jumper wires glow

Section B – True or False

1. An LDR can detect changes in light.....

2. The Arduino Uno is used to control the LED.....

3. The LED detects the brightness of the room.....